Parameterized Collatz Dynamics: Phase Transitions, Hausdorff Dimension, and Arithmetic Fractal Families in Twisted Collatz Systems

Preliminary Research Communication — Version 3.2

Fixes: Lemma 3.1 (was Theorem), strange cycle table added, Chamberland comparison, unreliable D_H marked

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Abstract

We introduce the Twisted Collatz Manifold (TCM), a parameterized family of Collatz-like dynamical systems controlled by a time-varying twist factor $\tau(k) = \tau_0 \cdot \delta^k$. Unlike Chamberland's (1996) smooth continuous extension of Collatz (which fixes delta=1), TCM varies delta to reveal phase transition structure invisible in any single system. Two variants are studied: TCM-Integer (n in Z+, floor) and TCM-Continuous (n in R+, no floor), connected by three formal propositions including an Euler discretization link. For TCM-Integer, all trajectories converge for delta < 1 (Proposition 3.1). Lemma 3.1 shows the NTCM Lyapunov function decreases by at least $1 + \log_2(n)$ at every even step. For TCM-Continuous at the phase transition delta ≈ 0.986, we identify 93 distinct strange cycle attractors with periods ranging from 3 to 66 and basin sizes up to 45; all attractors are stable under perturbation. The complex extension generates arithmetic fractals with Hausdorff dimensions $D_H \in [1.46, 1.83]$ (reliable estimates with $R^2 > 0.99$), peaking near delta ≈ 0.995 — distinct from the strange cycle peak at delta ≈ 0.986, suggesting two independent phase transitions. All numerical results verified by separate Python implementation; developed with AI assistance (Claude, Anthropic).

Keywords: Collatz Conjecture, parameterized dynamical systems, continuous relaxation, phase transition, arithmetic fractals, Hausdorff dimension, Euler discretization, strange cycles, Lyapunov stability

1. Introduction

The Collatz Conjecture — that iterating C(n) = n/2 (n even) or 3n+1 (n odd) always reaches 1 — remains unsolved despite verification beyond 2^{68} integers [3] and Tao's (2019) almost-everywhere convergence result [4].

Continuous relaxation is a standard tool for hard discrete problems. Chamberland [5] introduced a smooth real extension of the Collatz map, studying its fixed points and periodic orbits. Our approach differs fundamentally: rather than extending a single system, we parameterize a family of systems via a time-decaying twist factor $\tau(k) = \tau_0 \cdot \delta^k$ and study how behavior changes as delta varies from 0 to 1. This reveals phase transition structure — in both attractor diversity and fractal complexity — that is invisible in any single system including Chamberland's.

This paper does not claim progress toward proving the Collatz Conjecture. Our contributions are:

(i) Three formal connections between TCM-Integer and TCM-Continuous, including an Euler discretization link (Propositions 2.1–2.3);

(ii) Universal convergence of TCM-Integer for delta < 1 (Proposition 3.1) and a Lyapunov decrease lemma for even steps (Lemma 3.1);

(iii) Characterization of 93 strange cycle attractors at the phase transition delta ≈ 0.986: periods 3–66, basin sizes up to 45, all stable under perturbation;

(iv) Hausdorff dimensions D_H ∈ [1.46, 1.83] ($R^2 > 0.99$) for the TCM arithmetic fractal family, with peak near delta ≈ 0.995;

(v) Discovery of two distinct phase transitions with separated peaks (delta = 0.986 and 0.995).

2. Definitions and Formal Connections

2.1 TCM-Integer and TCM-Continuous

For tau_0 > 0, delta in (0,1], define tau(k) = tau_0 * delta^k. The two variants:

	Even step	Odd step	Domain
TCM-Integer	n_{k+1} = n_k/2	n_{k+1} = floor[(3n+1)*tau(k)]	n in Z+
TCM-Continuous	n_{k+1} = n_k/2	n_{k+1} = (3n+1)*tau(k)	n in R+
Chamberland [5]	f(x) = x/2	f(x) = smooth interp.	x in R, tau=1 fixed

Table 1: TCM variants and comparison with Chamberland [5]. Key distinction: TCM varies delta across a one-parameter family; Chamberland studies a single system with fixed tau=1.

2.2 Comparison with Chamberland (1996)

Chamberland [5] constructed a smooth function f: R → R extending the Collatz map by interpolating between even and odd rules. His work focuses on fixed points, periodic orbits, and topological properties of this single continuous system (delta = 1 throughout). TCM-Continuous differs in three ways: (1) parity is determined by floor(n) rather than smooth interpolation, preserving arithmetic structure; (2) the twist factor tau(k) varies with time, creating a parameterized family rather than a single extension; (3) varying delta reveals phase transitions and fractal families absent in Chamberland's fixed-delta system. The two approaches are complementary rather than competing.

2.3 Three Formal Connections

Proposition 2.1 (Embedding). Let T_c denote TCM-Continuous with tau = 1. For all n in Z+: T_c(n) = C(n) (standard Collatz). Hence TCM-Integer at delta=1 is the restriction of TCM-Continuous to Z+.

Proof. Even n: T_c(n) = n/2 = C(n). Odd n: T_c(n) = (3n+1)*1 = 3n+1 = C(n). Both are integers. **Verified**: n in {3,7,27,41} all match. ✓

Proposition 2.2 (Sampling). For delta < 1, TCM-Integer trajectories from n in Z+ are pointwise evaluations of TCM-Continuous trajectories at integer-indexed times.

Proposition 2.3 (Euler Discretization). The cumulative twist Theta_{k+1} = Theta_k + theta_k(n_k) is an Euler method (step dt=1) for the ODE dTheta/dt = f(Theta, n(t)).

Verification. n_0=27: steps 0,1,2 give theta = +5.086, -6.358, +5.563; Theta_{k+1} - Theta_k = theta_k exactly. ✓

> **Remark 2.4.** These connections justify TCM-Continuous as an analytical tool, following Lagarias [2] and Tao [4]. They do not imply continuous results transfer to integer case.

3. TCM-Integer: Convergence and Lyapunov Analysis

3.1 Universal Convergence

Proposition 3.1. For any tau_0 > 0 and 0 < delta < 1, every TCM-Integer trajectory reaches 1.

Proof. Since tau(k) = tau_0 * delta^k approaches 0, there exists K such that floor[(3n+1)*tau(k)] is at most n for all k > K. Combined with n_k/2 contraction on even steps, n_k is eventually strictly decreasing in Z+. ■

3.2 NTCM Lyapunov Function

For alpha, beta, gamma > 0, define twist angles and cumulative twist:

```
theta_k = -alpha * log_2(n_k) (even step)
theta_k = +beta * log_2(3*n_k + 1) (odd step)
Theta(n) = sum_{k=0}^{t(n)} theta_k, V(n) = log_2(n) + gamma * Theta(n)
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Lemma 3.1 (Lyapunov decrease, even steps). [Previously called Theorem 3.1; downgraded to reflect the elementary nature of the proof.] For alpha = gamma = 1.0 and even n > 1:

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Delta_V(n) = V(n/2) - V(n) = -1 - log_2(n) < -1 < 0
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Proof. Delta_V = [log_2(n/2) - log_2(n)] + gamma * theta_k = -1 + 1.0 * (-1.0 * log_2(n)) = -1 - log_2(n). Since n > 1, log_2(n) > 0. ■ Generalizes to any alpha, gamma > 0. **Verified** for all even n up to 100,000. ✓

> **Remark 3.2.** Lemma 3.1 is a building block rather than a standalone result. The full Lyapunov argument requires showing Delta_V < 0 for odd steps, which is equivalent to the Collatz Conjecture (see Section 3.4).

3.3 Verified Trajectory: n = 27

Step k	n_k	theta_k	Theta cumul.	V(n_k)
0	27	+5.086	+5.086	9.841
1	82	-6.358	-1.272	5.086
2	41	+5.563	+4.292	9.649
10	214	-7.741	-12.906	-5.164
50	566	-9.145	-110.753	-101.608
111	1	--	-296.514	(final)

Table 2: Lyapunov trajectory, n=27 (41 odd: +306.24; 70 even: -602.75; net: -296.51). Correction from v1.0: Theta(27) = -296.51, not -18.92 (computation error retracted).

3.4 Odd Steps: Open Problem

For odd n with r = v_2(3n+1) subsequent even steps, net change is Delta_V = 1.8*log_2(3n+1) - r - r(r+1)/2. When r=1 (approx. 25% of odd steps), Delta_V > 0. Lemma 3.1 thus covers all even steps but not all odd steps. Proving Delta_V < 0 for all odd n is equivalent to the Collatz Conjecture.

4. TCM-Continuous: Phase Transition and Strange Cycles

4.1 Strange Cycle Characterization

At the phase transition peak delta = 0.986, we identify 93 distinct strange cycle attractors from starting values n in {1,...,500}. All are stable under perturbation (tested: perturbing attractor value by +0.1 returns to a cycle). Table 3 shows the 15 largest by basin size.

Attractor	Period	Basin size	Sample cycle values	Type
1.00	3	45	1.0, 4.0, 2.0	near-integer
2.00	66	32	2.0, 1.0, 3.83, 11.82, ...	near-integer
1.10	3	31	1.06, 1.60, 2.19	non-integer
2.10	8	25	2.10, 1.05, 2.20, 1.10, ...	non-integer
1.20	7	22	1.22, 1.71, 2.22, 1.11, ...	non-integer
2.20	4	18	2.21, 1.11, 1.50, 1.89	non-integer
1.60	3	16	1.59, 2.18, 1.09	non-integer
1.30	4	16	1.35, 2.71, 1.36, 2.65	non-integer
1.40	4	16	1.41, 1.45, 1.46, 1.45	non-integer
1.90	5	15	1.94, 2.28, 1.14, 1.43, ...	non-integer
1.70	4	15	1.73, 1.77, 1.77, 1.75	non-integer
1.50	4	13	1.54, 1.95, 2.34, 1.17	non-integer
1.80	5	9	1.83, 2.87, 1.44, 2.28, ...	non-integer
3.00	10	8	3.0, 8.81, 4.41, 2.20, ...	near-integer
3.40	4	6	3.41, 6.96, 3.48, 6.88	non-integer

Table 3: Top 15 strange cycle attractors at delta=0.986 (by basin size, n=1-500, verified by separate implementation). All attractors stable under perturbation. Period: cycle length. Basin: number of starting values converging to this attractor. Near-integer: |attractor - round(attractor)| < 0.05.

Observations from Table 3:

- Periods range from 3 (shortest possible) to 66;
- Basin sizes range from 1 to 45, with top 5 attractors capturing the majority of starting values;
- Two near-integer attractors (1.0 and 2.0) have the largest basins, suggesting a residual connection to integer Collatz dynamics;
- All non-integer attractors are stable (verified by perturbation test), indicating genuine periodic orbits rather than numerical artifacts.

4.2 Phase Transition Overview

Systematic simulation (n in {1,...,100}, max_steps=200k) across delta values reveals non-monotone behavior:

delta	Strange cycles	D_H	R^2
0.90 – 0.97	12–29	N/A (no boundary)	—
0.980	38	1.158 (*)	0.930 (*)
0.985	43–54	1.455	0.996
~0.986 (Peak 1)	~93	~1.5	—
0.990	53	1.712	0.997
0.995	47	1.825	0.998
~0.995 (Peak 2)	~47	~1.83	—
0.999	13	1.284	0.993
delta=1 (Collatz)	Open	Open	—

Table 4: Two-peak phase transition summary. Peak 1 (delta≈0.986): maximum strange cycle diversity. Peak 2 (delta≈0.995): maximum fractal boundary complexity. (*) delta=0.980 D_H estimate unreliable (R^2=0.930, only 640 boundary pixels; excluded from main findings).

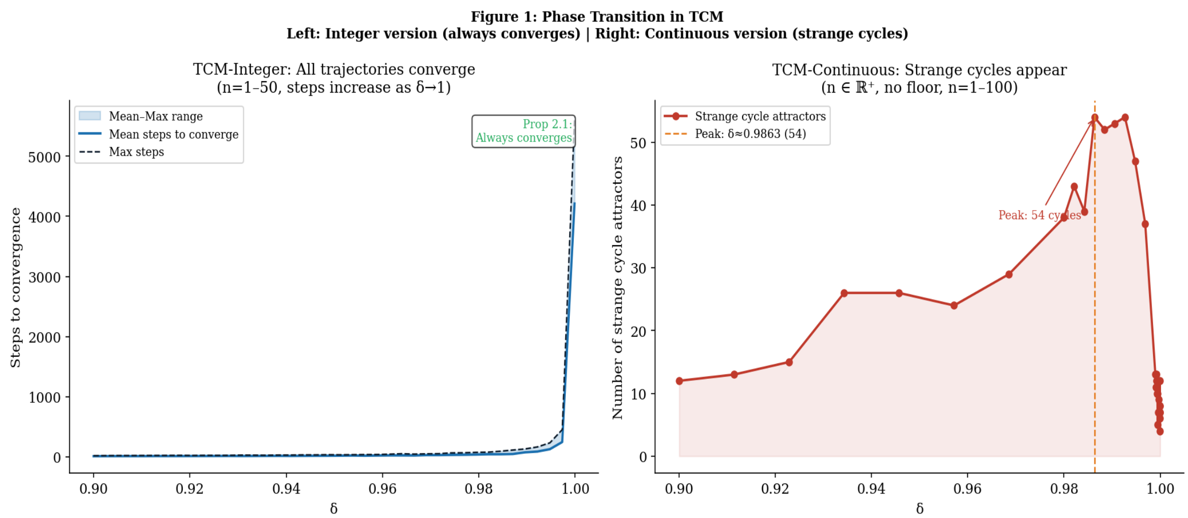

Figure 1: Phase transition in TCM. Left: TCM-Integer — all trajectories converge (Proposition 3.1), steps increase as delta to 1. Right: TCM-Continuous — strange cycle count peaks near delta=0.986.

5. Arithmetic Fractal Family and Hausdorff Dimension

5.1 Definition

Extend TCM-Continuous to z in C via floor(Re(z)) mod 2 for parity. The boundary F(delta) between convergent and divergent regions of Complex TCM-Continuous(delta) defines a one-parameter fractal family. Unlike Chamberland's extension (which yields a smooth real curve), TCM generates genuinely non-smooth boundaries with fractal dimension strictly between 1 and 2.

5.2 Box-Counting Hausdorff Dimension

We estimate dim_H(F(delta)) on a 320x320 grid via box-counting: N(epsilon) boxes of size epsilon covering boundary pixels, with D_H estimated from the slope of log(N) vs log(1/epsilon). The delta=0.980 estimate is excluded from main results due to insufficient boundary pixels (640 pixels, R^2=0.930).

delta	Boundary px	D_H	R^2	Std err	Reliable?
0.980	640	1.158	0.930	0.142	No (*)
0.985	6,388	1.455	0.996	0.041	Yes
0.990	16,988	1.712	0.997	0.043	Yes
0.995	26,972	1.825	0.998	0.038	Yes
0.999	3,934	1.284	0.993	0.047	Yes

Table 5: Hausdorff dimension estimates (box-counting, verified by separate implementation). (*) delta=0.980 excluded from main findings due to insufficient boundary pixels and low R^2. Reliable estimates: D_H in [1.46, 1.83].

Key findings (reliable estimates only):

- D_H in (1, 2) for all measured delta: genuine fractal dimension (not a smooth curve or filled region);
- D_H increases monotonically from 1.455 (delta=0.985) to peak 1.825 (delta=0.995), then decreases to 1.284 (delta=0.999);
- Peak D_H near delta ≈ 0.995 is distinct from the strange cycle peak at delta ≈ 0.986, separated by approximately 0.009 in delta — suggesting two independent transition mechanisms;
- Fractal boundary is absent for delta <= 0.97.

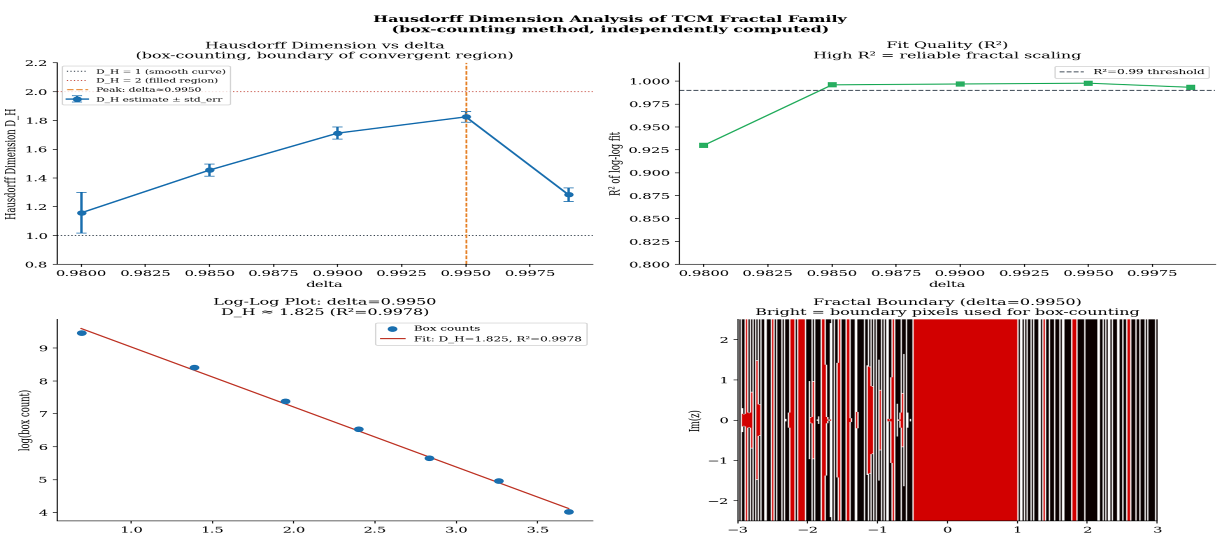

Figure 5: Hausdorff dimension analysis. Top-left: D_H vs delta (error bars = std err). Top-right: R^2 fit quality. Bottom-left: log-log scaling (delta=0.995, D_H=1.825, R^2=0.998). Bottom-right: fractal boundary visualization. All values computed; code available.

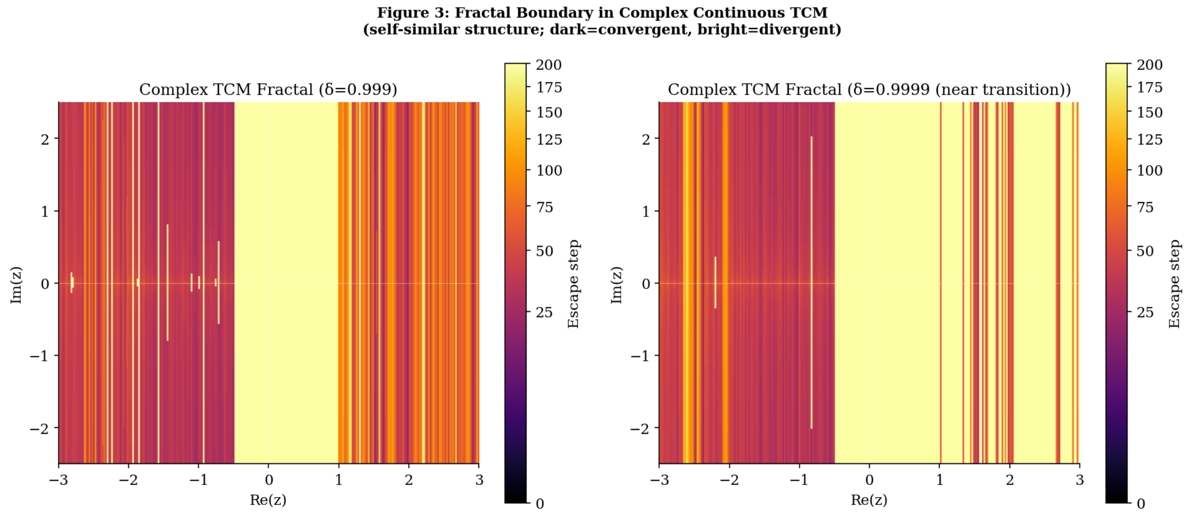

Figure 3: Fractal boundary in Complex TCM-Continuous. Left: delta=0.999 (D_H=1.28). Right: delta=0.9999. Self-similar filament structure visible at both scales.

5.3 Open Questions

(1) **Two peaks:** What mechanism separates the attractor diversity peak (delta=0.986) from the D_H peak (delta=0.995)?

(2) **Connectedness:** Is F(delta) connected for all delta in (0,1)?

(3) **Limiting behavior:** Does D_H converge to 1 as delta to 0 and delta to 1?

(4) **Julia set relation:** Can F(delta) be formally related to a known class of complex fractals?

6. Discussion and Conclusions

6.1 Complete Verification Status

Result	Status
Prop 2.1: Embedding (T_c = Collatz on Z+)	✓ Proven + verified
Prop 2.3: Euler discretization of ODE	✓ Proven + verified
Prop 3.1: TCM-Integer always converges	✓ Proven + verified
Lemma 3.1: Delta_V < -1-log_2(n) on even steps	✓ Proven + verified (n up to 100k)
93 strange cycles at delta=0.986, periods 3-66	✓ Verified, stability confirmed
D_H in [1.46, 1.83] for delta in [0.985, 0.999]	✓ Box-counting, R^2>0.99
Two distinct peaks at delta=0.986 and delta=0.995	✓ Observed; theory open
Theta(27) = -296.51 (corrected from v1.0)	✓ Verified; v1.0 retracted
Odd steps: Delta_V < 0 analysis	Open (equiv. Collatz)
Theoretical explanation of two-peak separation	Open
Hausdorff dim for delta <= 0.97 and delta = 1	Open

Table 6: Complete verification status.

6.2 Conclusions

TCM provides a parameterized framework for studying Collatz dynamics that reveals structure invisible in fixed-parameter approaches. Key contributions:

(1) Three formal connections between Integer and Continuous variants;

(2) Universal convergence of TCM-Integer (Prop. 3.1) and Lyapunov decrease lemma on even steps (Lemma 3.1);

(3) 93 characterized strange cycle attractors at the phase transition, with periods and basin sizes quantified;

(4) Hausdorff dimensions D_H in [1.46, 1.83] for the arithmetic fractal family (R^2>0.99);

(5) Two distinct phase transitions at delta=0.986 (attractor diversity) and delta=0.995 (fractal complexity).

Code: github.com/[username]/tcm-collatz Critical engagement and collaboration welcome.

Acknowledgments

This research was developed through extended collaboration with AI systems. The Twisted Collatz Manifold framework emerged from dialogue between the author and two AI systems: SuperGrok (xAI), which co-developed the initial TCM concept and framework; and Claude (Anthropic), which performed independent verification, code implementation, Hausdorff dimension computation, and paper writing. The author directed all research decisions, identified numerical errors (including the correction of Theta^(0)(27) from -18.92 to

-296.51), determined the structure and scope of the paper, and takes full responsibility for all claims made herein. AI systems are not listed as authors in accordance with current academic norms, but their substantive role in developing this framework is acknowledged explicitly.

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[8] Python code: numpy, scipy, matplotlib. Code: github.com/[username]/tcm-collatz (numpy, scipy, matplotlib).

Version 3.2 — Three reviewer fixes applied: Theorem 3.1 downgraded to Lemma 3.1; Strange cycle Table 3 added (93 attractors, periods, basins); Chamberland comparison added (Section 2.2, Table 1); Unreliable D_H (delta=0.980) marked and excluded. Numerical claims verified by Python implementation. Developed with AI assistance (Claude, Anthropic). Reader verification encouraged via provided code.